Consolidated Computational Analysis: Dual-Compartment Bacteremia Model, Monte Carlo Procedure, and Deterministic Switch Localization

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Contents

1 Purpose and scope **2**

2 Model and treatment protocol **2**
2.1 State variables . . . 2
2.2 Drug-effect functions . . . 3
2.3 Centralized configuration . . . 3

3 Vancomycin-sensitive-strain clearance **3**

4 Numerical integration and the hard LOD **4**
4.1 The hard LOD is a modeling assumption . . . 5

5 Event location at the limit of detection **5**

6 Two related but distinct analyses **6**

7 Determination of transition points with Brent's method **6**
7.1 The quantity being localized . . . 6
7.2 What the sign does—and does not—mean . . . 6
7.3 How the bracket is found . . . 7
7.4 What Brent's algorithm does . . . 7
7.5 How this differs from event location . . . 8
7.6 Important limitation: this is a discontinuous boundary . . . 8
7.7 Current deterministic switches . . . 9

8 Monte Carlo procedure **9**
8.1 Meaning of a Monte Carlo iteration . . . 9
8.2 Why a mean-corrected log-normal distribution is used . . . 9
8.3 Exact computational sequence . . . 10
8.4 Clinical outcome definitions . . . 11
8.5 Trajectory plots and missing-value display convention . . . 11

9 From switch distance to relapse probability **11**
9.1 Why the raw distance is insufficient 11
9.2 Choosing the correct tail 12
9.3 Current distance and outcome calculations 12

10 Analysis-specific interpretation and verified results **12**

11 Reproducibility map **13**

12 Interpretive cautions **14**

1 Purpose and scope

This document is the canonical, standalone explanation of analyses previously divided among five focused notes, which were removed after consolidation:

- `EC50_lzd_analysis.tex`;
- `ec50_threshold_comparison_summary.tex`;
- `MC_immune_response_procedure.tex`;
- `MC_scripts_comparison.tex`; and
- `vancomycin_sensitive_clearance.tex`.

Obsolete descriptions from those notes are not carried forward. In particular, the current model uses `solve_ivp`, not `odeint`; uses the current centralized growth rates; and treats the limit of detection as a hard extinction/reactivation boundary.

Three one-parameter Monte Carlo analyses are included:

1. linezolid half-maximal concentration, $EC_{50,L}$;
2. linezolid maximum killing effect, $E_{\max,L}$; and
3. host immune-clearance rate, k_{immune}.

Each analysis varies only the named parameter. It is therefore a univariate sensitivity/variability analysis, not a joint population model in which all parameters vary simultaneously. Initial conditions are fixed and are not an analysis variable.

2 Model and treatment protocol

2.1 State variables

The state vector is

$$\mathbf{Y}(t) = \big[S_b(t), R_b(t), S_{\text{res}}(t), R_{\text{res}}(t)\big], \tag{1}$$

where S and R denote vancomycin-sensitive and vancomycin-resistant bacteria, and the subscripts denote blood and the bone reservoir. The fixed starting state is

$$\mathbf{Y}(0) = [0, 0, 100, 100] \text{ CFU/mL}. \tag{2}$$

This starting state is required to run the model but is not sampled in the present analyses.

The simulation lasts 1944 h (81 days): 21 days before treatment, 4 days of vancomycin, 42 days of linezolid, and 14 days of post-linezolid follow-up.

2.2 Drug-effect functions

With Hill coefficients equal to one, the blood-compartment effects are

$$E_V(t) = \frac{E_{\max,V} V(t)}{EC_{50,V} + V(t)}, \tag{3}$$

$$E_L(t) = \frac{E_{\max,L} L(t)}{EC_{50,L} + L(t)}. \tag{4}$$

Larger $EC_{50,L}$ means weaker linezolid potency at the same concentration, whereas larger $E_{\max,L}$ means a larger maximum killing capacity. Reservoir drug concentrations are the corresponding blood concentrations multiplied by the centralized tissue-penetration fractions. In the reservoir the linezolid ceiling is further reduced by an independent efficacy factor, $E^{\text{res}}_{\max,L} = \alpha\, E_{\max,L}$ with $\alpha = 0.45$, representing reduced activity against slow-growing biofilm and small-colony-variant bacteria in bone; α is calibrated and is not derived from the growth rates.

2.3 Centralized configuration

All scripts obtain the fixed configuration from `parameters.py`. Values are shown to two significant figures.

Quantity	Value	Interpretation
ρ_S	1.47 h^{-1}	sensitive growth, blood (Vera-Yunca et al., in vitro)
ρ_R	1.323 h^{-1}	resistant growth, blood
$\rho_{\text{res},S}$	0.845 h^{-1}	sensitive growth, reservoir (Vera-Yunca et al., in vivo)
$\rho_{\text{res},R}$	0.761 h^{-1}	resistant growth, reservoir
$E_{\max,V}$	1.7 h^{-1}	maximum vancomycin effect
$EC_{50,V}$	0.25 mg/L	vancomycin half-maximal concentration
$E_{\max,L}$	1.97 h^{-1}	linezolid maximum effect (Vera-Yunca et al.)
$EC_{50,L}$	0.56 mg/L	linezolid half-maximal concentration (Vera-Yunca et al.)
$B_{\max,b}$	6.0×10^3 CFU/mL	blood carrying capacity
$B_{\max,\text{res}}$	1.0×10^7 CFU/mL	reservoir carrying capacity
ϕ_V	0.15	reservoir vancomycin penetration
ϕ_L	0.50	reservoir linezolid penetration
α	0.45	reservoir linezolid efficacy factor
$k_{r\to b}$	1.0×10^{-7} h^{-1}	reservoir-to-blood transfer
k_{immune}	0.18 h^{-1}	baseline immune clearance
ϵ_{res}	0.50	reservoir immune multiplier
L_{OD}	10 CFU/mL	hard detection boundary

Table 1: Current shared configuration. The resistant-strain fitness cost is 10% in both blood and the reservoir. Exchange is one-way, reservoir to blood; there is no blood-to-reservoir term.

3 Vancomycin-sensitive-strain clearance

Vancomycin acts directly on S_b and S_{res} but not on the resistant populations. At low blood burden and saturating vancomycin, the approximate per-capita sensitive-blood rate is

$$\rho_S - k_{\text{immune}} - E_{\max,V} = 1.47 - 0.18 - 1.7 = -0.41 \text{ h}^{-1}. \tag{5}$$

The negative value explains the rapid decline of the sensitive strain after vancomycin begins. Removal of sensitive bacteria also releases shared carrying capacity, allowing resistant bacteria to expand during vancomycin treatment.

The current baseline hybrid simulation gives the following values. A zero is an exact model state produced by the hard-LOD rule, not an arbitrarily small positive value rounded for display.

Time	S_b	R_b	S_{res}	R_{res}
Vancomycin start	4.6×10^3	0	8.0×10^6	3.1×10^4
+24 h	0	5.1×10^3	0	8.8×10^6
+48 h	0	5.2×10^3	0	8.8×10^6
+72 h	0	5.2×10^3	0	8.8×10^6
Vancomycin end	0	4.4×10^3	0	8.3×10^6

Table 2: Current baseline burdens during the four-day vancomycin course, in CFU/mL.

Thus vancomycin clears both sensitive compartments within the first day of treatment under the current calibration. Resistant blood bacteria are undetectable before treatment and become detectable once vancomycin frees the shared carrying capacity. Subsequent linezolid treatment is responsible for control of the resistant populations.

4 Numerical integration and the hard LOD

All current analyses call `solver.integrate_model`, which wraps `scipy.integrate.solve_ivp`. The configured method is LSODA, with relative tolerance 10^{-6}, absolute tolerance 10^{-8} CFU/mL, and maximum internal step 4.0 h. The canonical output grid starts from 1450 evenly spaced values and inserts the three treatment boundaries, producing 1453 output times.

The solver is hybrid because it combines continuous ODE integration with discrete state changes:

1. While a population is detectable, it follows the continuous ODE.
2. A downward event at $L_{\text{OD}} = 10$ sets that population to exactly zero.
3. At zero, the population has no autonomous growth, drug killing, immune clearance, or outgoing transfer.
4. Positive incoming transfer from a detectable paired compartment is allowed to accumulate.
5. When accumulated arrivals reach the LOD, an upward event reactivates the population and continuous dynamics resume.

This construction prevents a fractional sub-LOD remnant from manufacturing a relapse while still permitting a persistent reservoir to reseed blood.

Three consequences of these rules affect how the results are read:

1. **Initial state.** The rule is applied at time zero. Any population that starts below the LOD is set to zero and begins extinct. Blood starts sterile, so both blood populations begin extinct, and bacteremia first appears only when seeding from the reservoir has accumulated to 10 CFU/mL.
2. **Reservoir extinction is permanent.** Transfer in the model is one-way, from reservoir to blood. Only a blood population can therefore be reactivated. An extinct reservoir population receives no incoming transfer and remains at zero for the rest of the simulation.

3. **Carrying capacity.** An extinct population, including any arrivals it is accumulating, does not count toward the shared carrying capacity of its compartment.

4.1 The hard LOD is a modeling assumption

Using a cutoff at all addresses a known defect of deterministic population models, in which a burden far below one organism can decay and later regrow [1, 2]. Three specific choices are assumptions of this model and are not taken from a cited source:

1. The cutoff is placed at the assay limit of detection, 10 CFU/mL, and not at one organism in the compartment volume. Populations below the cutoff are treated as eradicated, not merely unobserved.

2. An extinct population returns only when transferred organisms accumulate to the LOD.

3. While they accumulate, those transferred organisms are protected. They are not subject to antibiotic killing, immune clearance, or any other loss, so the accumulated amount can only increase, and it is retained even if the donor population later becomes extinct. This favors reseeding: arrivals that treatment or the immune response might have removed are all counted toward reactivation.

5 Event location at the limit of detection

Extinction and reactivation at the LOD are handled inside each simulation by event location. This procedure acts on *time*, within a single simulation. It is separate from the determination of the transition points in Section 7, which acts on a *parameter*, across simulations.

The crossings are found with the event detection built into `solve_ivp` [3]. Each of the four populations y_i is given an event function

$$g_i(t) = y_i(t) - L_{\mathrm{OD}}. \tag{6}$$

1. **Direction.** A detectable population is watched only for a downward crossing of g_i (extinction). An extinct population is watched only for an upward crossing (reactivation).

2. **Detection.** `solve_ivp` compares the sign of g_i at the two ends of every integration step. A crossing is noticed only when the sign differs between them.

3. **Location in time.** When g_i changes sign within an integration step, `solve_ivp` applies `brentq` to g_i over that step, evaluating the solution with the integrator's continuous interpolant. The tolerance is four units of machine precision (SciPy 1.17.1, `scipy/integrate/_ivp/ivp.py`, `solve_event_equation`). Because g_i is continuous in time and changes sign over the step, a true root exists.

4. **Reset and restart.** Every event is terminal. Integration stops at the crossing time; an extinct population is set to exactly zero and a reactivated one is restarted at $L_{\mathrm{OD}}(1 + 10^{-6})$, a small offset that keeps it from re-triggering the event immediately. Integration then resumes from that time. The trajectory is therefore a sequence of continuous segments separated by discrete resets [4, 5].

5. **Safeguard.** More than 10 000 events in one simulation raises an error instead of returning a trajectory.

Only LOD crossings are events. Dose administration times and the change from vancomycin to linezolid are not; they are resolved by the 4.0 h maximum step. The event times themselves are not returned: the solver reports burdens on the output grid only, with every value below the LOD reported as zero. The arrivals accumulating in an extinct population are therefore invisible in the output until they reach the LOD.

The solver does not return a trajectory it cannot vouch for. An integrator warning, a failed step, or a non-finite value raises an error, and an initial state that is negative or non-finite is rejected before integration begins.

Event location therefore determines when, within one simulation, each population becomes extinct or is re-established. It does not determine the value of any parameter.

6 Two related but distinct analyses

The deterministic and Monte Carlo calculations answer different questions.

Deterministic switch analysis. If every value except one parameter is fixed, where does the model change between final reservoir clearance and final reservoir persistence?

Monte Carlo analysis. If that parameter differs among 1000 virtual patients according to a specified distribution, what fraction experiences bacteremia relapse, persistent bacteremia, or resolution?

The deterministic switch helps explain the Monte Carlo percentage, but it does not assign the clinical outcome. Clinical outcomes are calculated from each patient's complete blood trajectory.

7 Determination of transition points with Brent's method

The manuscript calls the located values *transition points*. This document also calls them *switches*; the two terms mean the same thing.

7.1 The quantity being localized

Let θ be the parameter being varied and let t_f be the end of the 81-day simulation. Define

$$F(\theta) = R_{\text{res}}(t_f; \theta) - L_{\text{OD}}. \tag{7}$$

The sign is interpreted as

$$F(\theta) < 0 \quad \Longrightarrow \quad \text{reservoir cleared,} \tag{8}$$

$$F(\theta) > 0 \quad \Longrightarrow \quad \text{reservoir detectable.} \tag{9}$$

Because the hybrid solver returns zero rather than a positive sub-LOD value, the clearance side generally has $F = -10$ rather than a value approaching zero smoothly.

7.2 What the sign does—and does not—mean

The function F contains the *reservoir* burden at the final observation, not the blood burden after linezolid. Its sign must therefore be read only as follows:

$$F(\theta) < 0 \implies R_{\text{res}}(t_f) < 10 \quad \text{(reservoir cleared at day 81)},$$
$$F(\theta) > 0 \implies R_{\text{res}}(t_f) > 10 \quad \text{(reservoir persists at day 81)}.$$

Neither statement, by itself, says whether bacteremia occurred after linezolid. In particular, a negative value of F does **not** mean that post-linezolid bacteremia occurred. A positive value of F also does **not** automatically mean relapse.

Two times and two compartments are involved:

Quantity	Time	Question answered
$R_{\text{res}}(t_f)$	day 81, end of follow-up	Is the reservoir still detectable?
$R_b(t_L)$ and $R_b(t > t_L)$	day 67 and follow-up	Did bacteremia clear and then return?

Here t_L is linezolid completion and t_f is the end of the 14-day follow-up. If one wished to describe only whether blood became detectable during follow-up, a separate quantity would be

$$H(\theta) = \max_{t_L < t \le t_f} R_b(t;\theta) - L_{\text{OD}}. \tag{10}$$

Then $H(\theta) \ge 0$ means that bacteremia was detectable at least once during follow-up. Even this is not sufficient by itself to define relapse: relapse additionally requires $R_b(t_L) < L_{\text{OD}}$, so that blood was cleared when linezolid ended before becoming detectable again.

In the current grids the reservoir and blood classifications coincide at t_f: at every grid point of all three sweeps, the resistant blood burden at t_f was detectable exactly when the reservoir burden was. This is an observed property of these runs, which is why F separates resolution of the bacteremia from no resolution in practice. It is not part of the definition of F.

7.3 How the bracket is found

A deterministic grid is evaluated first. Adjacent values $a < b$ are selected when their classifications differ, equivalently when

$$F(a)F(b) < 0. \tag{11}$$

This establishes that the switch lies somewhere inside $[a, b]$, provided the classification changes only once within that interval. The grid is therefore used to establish direction, monotonicity, and a safe bracket; its midpoint is not accepted as the final switch estimate.

The two groups of scripts bracket differently. The Monte Carlo scripts pass the adjacent grid pair to `brentq`, and only when the grid shows exactly one change. The dedicated deterministic sweep scripts check that the two ends of the grid have opposite classifications and pass the whole range. Both satisfy Eq. (11), and they return the same switch to within the tolerance (for example 0.76202 and 0.76217 mg/L for $EC_{50,L}$).

7.4 What Brent's algorithm does

The bracket is passed to SciPy's `scipy.optimize.brentq` implementation [6]. Brent's method combines three ideas [7]:

1. **Bisection** tries the midpoint of the bracket. It is slow but reliable because it always reduces a valid sign-changing interval.

2. **Secant interpolation** uses the two endpoint values to estimate where a zero would lie and is often faster than bisection.

3. **Inverse quadratic interpolation** uses three recent evaluations to make an even faster estimate when their geometry is suitable.

At each iteration, the algorithm accepts an interpolation step only when it is judged safe; otherwise it falls back to bisection. After evaluating the new point, it discards the half of the interval that does not contain the sign change. Repetition contracts the bracket until its width satisfies the requested absolute parameter tolerance.

The absolute tolerance is 10^{-3} for $EC_{50,L}$ (mg/L) and $E_{\max,L}$ (h^{-1}), and 10^{-6} h^{-1} for k_{immune}, in both the deterministic sweep script and the Monte Carlo script for each parameter. The search stops when the width of the bracket falls below the tolerance, so the reported value lies within the tolerance of the switch. Transitions are reported to two significant figures, except $E_{\max,L}$, which is reported to three (1.85 h^{-1}) because its narrow sampling distribution makes the difference between 1.85 and 1.9 material.

7.5 How this differs from event location

Determining a transition point is distinct from the event location of Section 5; Table 3 compares them. Event location finds a time within one simulation and is followed by a change of the equations and a restart of the integration. The transition search finds a parameter value across simulations: each evaluation of F is one complete simulation, including all of its event locations, and nothing is switched or restarted.

The two procedures share only the root-finding algorithm. `solve_ivp` calls `brentq` on g_i internally; `solver.py` never calls it. The analysis scripts call `brentq` on F explicitly. They differ in what the algorithm can deliver: g_i is continuous in time, so a root is found, whereas F is discontinuous in θ, so the search localizes a change of outcome and not a root. The two are connected in one direction only: the reset to zero at the LOD within each simulation is what makes F discontinuous.

	Event location	Transition point
Scope	Within one simulation	Across simulations
Unknown	Time of an LOD crossing	Parameter value θ
Function	$g_i(t)$, Eq. (6)	$F(\theta)$, Eq. (7)
Continuity	Continuous in t	Discontinuous in θ
Result	A root, $g_i = 0$	Location of a change of sign
Action that follows	Equations switched, integration restarted	None; the value is reported
Who calls `brentq`	`solve_ivp`, internally	The analysis scripts

Table 3: Event location at the limit of detection compared with the determination of a transition point. Both use Brent's method, but for different purposes.

7.6 Important limitation: this is a discontinuous boundary

The SciPy documentation specifies continuity as an assumption for `brentq` [6]. Equation (7) is not continuous under the hard-LOD rule. On the clearance side the reservoir burden at t_f is exactly zero, so $F = -L_{\text{OD}}$; on the persistence side the reservoir has regrown toward its carrying capacity and F is of order 10^6 to 10^7 CFU/mL. F takes no value in between, so no θ in the bracket satisfies

$F(\theta) = 0$. Consequently, the usual theorem that Brent's method converges to a root satisfying $F(\theta) = 0$ does not apply here.

The method is still applicable. Its termination does not depend on continuity: the bisection safeguard reduces the bracket to the requested width while F keeps opposite signs at the two endpoints [8, 7]. Continuity is needed only for the further conclusion that the final bracket contains a zero.

In this analysis, `brentq` is used as a bracket-contraction procedure that numerically localizes a binary clearance–persistence change. The result should be called a *classification boundary*, *regime switch*, or *persistence threshold*; it should not be described as a conventional continuous root.

Nor has the switch been demonstrated to be a transcritical bifurcation. A transcritical bifurcation requires two equilibrium branches to meet and exchange stability. Establishing that claim would require equilibrium continuation and a Jacobian/eigenvalue stability analysis of the underlying continuous ODE. The present calculation instead uses a finite treatment course, a finite observation time, and discrete LOD events.

7.7 Current deterministic switches

Parameter	Grid	Points	Persistence side	Reported switch
$EC_{50,L}$	0.05–3.3 mg/L	131	above	0.76 mg/L
$E_{\max,L}$	0.70–2.2 h^{-1}	111	below	1.85 h^{-1}
k_{immune}	0.02–0.65 h^{-1}	127	below	0.093 h^{-1}

Table 4: Current resistant-reservoir classification boundaries. The unrounded computational values, rather than the displayed values, are used in subsequent probability calculations.

8 Monte Carlo procedure

8.1 Meaning of a Monte Carlo iteration

One iteration represents one virtual patient. The patient receives one random value of the parameter under study, while the treatment protocol, starting state, solver settings, and every other biological parameter remain fixed. The model is then integrated for the entire 81-day course. Repeating this procedure 1000 times produces a distribution of complete patient trajectories and outcomes.

This is parameter-variability propagation: the random number does not add noise to the ODE at each time step. Once a patient's parameter value is drawn, that value remains constant for the patient's entire simulation.

8.2 Why a mean-corrected log-normal distribution is used

All three sampled parameters must be positive. They are drawn from log-normal distributions. If X has required arithmetic mean m and coefficient of variation CV, the code calculates

$$\sigma_{\ln} = \sqrt{\ln(1 + CV^2)}, \tag{12}$$

$$\mu_{\ln} = \ln(m) - \frac{1}{2}\sigma_{\ln}^2, \tag{13}$$

$$X \sim \text{LogNormal}(\mu_{\ln}, \sigma_{\ln}^2). \tag{14}$$

The subtraction of $\sigma_{\ln}^2/2$ is the mean correction. It ensures

$$\mathbb{E}[X] = \exp\left(\mu_{\ln} + \frac{\sigma_{\ln}^2}{2}\right) = m. \tag{15}$$

Without this correction, entering $\ln(m)$ as the log-scale location would make m the median, not the mean, and the Monte Carlo calibration would no longer match the fixed baseline.

Analysis	Arithmetic mean	CV	Random seed
$EC_{50,L}$	0.56 mg/L	0.60	44
$E_{\max,L}$	1.97 h^{-1}	0.080	44
k_{immune}	0.18 h^{-1}	0.80	42

Table 5: Centralized Monte Carlo distributions. Each analysis contains 1000 draws and varies only the listed parameter.

8.3 Exact computational sequence

Each Monte Carlo script follows the same essential sequence:

1. Load the model, centralized parameters, canonical time grid, hybrid solver, and outcome-classification function.
2. Construct the vancomycin and linezolid concentration functions once.
3. Read the selected parameter's arithmetic mean, CV, and random seed from `parameters.py`.
4. Convert the mean and CV to $\mu_{\ln}$ and $\sigma_{\ln}$ using Eq. (14).
5. Initialize a dedicated NumPy random-number generator and draw all 1000 parameter values. Drawing them once makes the cohort reproducible.
6. Pre-allocate four arrays with 1000 rows and 1453 columns, one array for each bacterial state. Row i will contain virtual patient i's trajectory.
7. For patient i, copy the fixed baseline and override only the sampled parameter. For the immune analysis, construct a new `ImmuneResponse` object containing the sampled k_{immune}.
8. Integrate the complete four-state model over all 1453 reporting times. The hybrid solver has already applied extinction and reactivation events.
9. Store all four trajectories, then repeat until all 1000 patients are complete.
10. Apply the same trajectory-based outcome definitions to every patient's resistant-blood history.
11. Summarize counts, percentages, distributions, and time courses, and write the figures with parameter-provenance metadata.

8.4 Clinical outcome definitions

Let t_L denote linezolid completion and let the subsequent interval be the 14-day follow-up. The three outcomes are mutually exclusive:

$$\text{relapse}: \quad R_b(t_L) < L_{\text{OD}} \quad \text{and} \quad \max_{t>t_L} R_b(t) \geq L_{\text{OD}}, \tag{16}$$

$$\text{persistent failure}: \quad R_b(t_L) \geq L_{\text{OD}}, \tag{17}$$

$$\text{resolution}: \quad R_b(t_L) < L_{\text{OD}} \quad \text{and} \quad R_b(t) < L_{\text{OD}} \text{ for every } t > t_L. \tag{18}$$

A detectable reservoir without a post-treatment return of bacteremia is not called relapse. Conversely, a reservoir can cause relapse by remaining detectable through treatment and reseeding blood after linezolid ends.

The switch threshold is not used to force an outcome label. Every label is calculated from that patient's simulated blood trajectory. This distinction is important near the switch and whenever blood remains detectable at the end of treatment. Section 7.2 explains why the sign of the reservoir-based Brent objective cannot be interpreted directly as the presence or absence of post-linezolid bacteremia.

8.5 Trajectory plots and missing-value display convention

The model stores extinct populations as exact zeros. A logarithmic axis cannot display zero, so plotting code replaces zeros with `NaN` only for display and NaN-aware summaries. This does not change the simulation or the outcome classification.

For the EC50 and E-max kinetics figures, the median and 5th–95th percentile band are calculated within the patients who relapsed, making the reseeding trajectory visible. The immune-response figure calculates NaN-aware summaries over all patients; at each time these are conditional on the compartment being detectable. These conditional summaries must not be interpreted as unconditional cohort quantiles in which extinct patients have a small positive burden.

9 From switch distance to relapse probability

9.1 Why the raw distance is insufficient

The simple difference $T - m$ is not comparable across parameters because the parameters have different units and different CVs. A difference of 0.10 is large for a narrow distribution and small for a broad distribution. Because the sampling distributions are log-normal, distance is measured after taking natural logarithms and is divided by the log-scale standard deviation.

For an unrounded deterministic switch T, define

$$z_T = \frac{\ln(T) - \mu_{\ln}}{\sigma_{\ln}}. \tag{19}$$

The z-value answers: "How many log-scale standard deviations is the switch from the center of the sampling distribution?" The standard normal cumulative distribution function

$$\Phi(z) = P(Z \leq z), \qquad Z \sim N(0,1), \tag{20}$$

converts that standardized distance into an area under the distribution.

9.2 Choosing the correct tail

For $EC_{50,L}$, high values weaken linezolid, so persistence is the upper tail:

$$P(\text{persistence side}) = P(EC_{50,L} > T) = 1 - \Phi(z_T). \tag{21}$$

For $E_{\max,L}$ and k_{immune}, low values weaken killing or host clearance, so persistence is the lower tail:

$$P(\text{persistence side}) = P(X < T) = \Phi(z_T). \tag{22}$$

For example, the immune switch has $z = -0.59$. Approximately 27.9% of a standard normal distribution lies to the left of -0.59. Therefore about 27.9% of the configured immune distribution is expected to lie in the weak- immunity persistence regime. The seeded finite cohort placed 278 of 1000 patients (27.8%) on that side; a finite random sample need not equal the theoretical percentage exactly.

9.3 Current distance and outcome calculations

Parameter	z_T	Theoretical tail	Sampled side	Relapse	Persistent
$EC_{50,L}$	0.83	20.2%	222 (22.2%)	220 (22.0%)	2 (0.2%)
$E_{\max,L}$	−0.74	23.1%	226 (22.6%)	226 (22.6%)	0 (0.0%)
k_{immune}	−0.59	27.9%	278 (27.8%)	278 (27.8%)	0 (0.0%)

Table 6: Relationship between the deterministic switch, probability mass on the persistence side, and trajectory-defined clinical outcomes. The z-values use unrounded computational switches.

For each analysis in Table 6, the number on the persistence side equals relapse plus persistent treatment failure. Thus the switch-tail area estimates the *relapse-prone* fraction, not necessarily the relapse rate itself. A persistence-side patient is counted as relapse only if blood is below the LOD when linezolid ends and later becomes detectable. If blood is already detectable when linezolid ends, the outcome is persistent treatment failure.

This explains why persistence-side fractions and relapse percentages can differ. In the current calibration they nearly coincide: every persistence-side patient in the E-max (226) and immune (278) analyses first cleared blood and then relapsed, and only 2 of the 222 EC50 persistence-side patients still had detectable blood at treatment completion.

10 Analysis-specific interpretation and verified results

Analysis	Relapse	Persistent failure	Resolution
$EC_{50,L}$	220 (22.0%)	2 (0.2%)	778 (77.8%)
$E_{\max,L}$	226 (22.6%)	0 (0.0%)	774 (77.4%)
k_{immune}	278 (27.8%)	0 (0.0%)	722 (72.2%)

Table 7: Verified mutually exclusive outcomes for 1000 virtual patients per analysis. Each row sums to 1000.

Linezolid EC50.

Increasing $EC_{50,L}$ reduces potency. The clearance–persistence switch is 0.76 mg/L, and the sampling distribution has arithmetic mean 0.56 mg/L and CV 0.60. The high-EC50 upper tail is therefore the persistence-prone group.

Linezolid E-max.

Decreasing $E_{\max,L}$ lowers the maximum killing capacity. The reported switch is 1.85 h^{-1}, and the sampling distribution has arithmetic mean 1.97 h^{-1} and CV 0.080. Because this distribution is narrow, a small raw separation between the mean and switch can still correspond to a substantial standardized distance and tail probability.

Immune response.

Decreasing k_{immune} weakens host clearance. The switch is 0.093 h^{-1}, while the distribution mean is 0.18 h^{-1} with CV 0.80. This broad distribution places a clinically important lower tail below the switch even though the arithmetic mean is about twice the reported switch.

All three relapse percentages fall within the prespecified 20–30% calibration range. They are outcomes of the chosen means, CVs, other fixed parameters, hard-LOD logic, and relapse definition; they should not be interpreted as independently validated clinical predictions.

11 Reproducibility map

File	Role
`parameters.py`	Centralized parameters, sampling distributions, seeds, time grid, outcome definitions, and solver configuration.
`model_Bacteremia.py`	Pharmacokinetics, immune-response object, and four-state ODE right-hand side.
`solver.py`	LSODA integration with hard-LOD extinction and transfer-driven reactivation events.
`EC50_lzd_threshold_sweep.py`	Deterministic EC50 switch.
`Emax_lzd_threshold_sweep.py`	Deterministic E-max switch.
`ImmuneResponse_threshold_sweep.py`	Deterministic immune switch.
`MC_ec50_lzd.py`	EC50 Monte Carlo analysis and figures.
`MC_emax_lzd.py`	E-max Monte Carlo analysis and figures.
`MC_immune_response.py`	Immune-response Monte Carlo analysis and figures.

The principal figures are:

- `mc_ec50_lzd_sweep_kinetics.png`, `mc_emax_lzd_sweep_kinetics.png`, and `mc_immune_response.png`;
- the three corresponding switch/population figures; and
- `distribution_to_switch.png` and `relapse_time_curve.png`.

12 Interpretive cautions

1. The three Monte Carlo analyses are one-at-a-time analyses. They do not represent correlations or simultaneous variability among EC50, E-max, and immune response.
2. The CVs are outcome-calibrated inputs. They are not claimed to be population estimates obtained from an independent clinical dataset.
3. A deterministic reservoir switch is mechanistic context, not the clinical relapse definition.
4. The hard LOD is a modeling rule. It changes the mathematics from a fully continuous ODE to a hybrid dynamical system.
5. Brent's continuity-based root guarantee does not apply to the hard-LOD objective. The returned value localizes a classification change.
6. Calling the switch a transcritical bifurcation would require a separate equilibrium and stability analysis that has not been performed.
7. Displayed parameters and thresholds are rounded, but probability calculations use unrounded computational values. This is especially important for the narrow E-max distribution.

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